

Course Code	AZ-900	L	T	P	C

COURSE DESCRIPTION

This course introduces foundational concepts of cloud computing and core Microsoft Azure services aligned with the AZ-900 certification exam. Learners explore cloud principles, Azure architecture, compute, storage, networking, identity, security, governance and cost management. The course prepares students for Microsoft Azure Fundamentals certification and establishes a strong foundation for advanced Azure and cloud certifications.

COURSE OBJECTIVES

The course should enable the student to:

1. Explain cloud computing concepts and service models.
2. Describe Azure architectural components and resource hierarchy.
3. Identify Azure compute, storage, networking and identity services.
4. Explain Azure security, governance and compliance mechanisms.
5. Analyze Azure pricing, cost management and deployment tools.

SYLLABUS

Unit 1: Cloud Computing Fundamentals

Cloud Computing Fundamentals, Benefits of Cloud Services, Shared Responsibility Model, Public Private Hybrid Cloud, IaaS PaaS SaaS, Consumption-Based Pricing, High Availability, Scalability, Elasticity, Reliability, Disaster Recovery Concepts

Unit 2: Azure Architecture and Compute Services

Azure Regions and Availability Zones, Datacenters, Resources and Resource Groups, Subscriptions and Management Groups, Resource Hierarchy, Azure Compute Services, Virtual Machines, App Services, Containers Overview

Unit 3: Networking, Storage and Identity Services

Virtual Networks and Subnets, VPN Gateway, ExpressRoute, Load Balancer, DNS, Azure Storage Types, Storage Redundancy, Azure SQL Database, Cosmos DB Overview, Microsoft Entra ID, Authentication Methods, Role-Based Access Control

Unit 4: Security, Governance and Compliance

Azure Security Concepts, Zero Trust Principles, Multifactor Authentication, Conditional Access, Azure Policy, Resource Locks, Tagging, Compliance Offerings, Microsoft Purview Overview

Unit 5: Pricing, Cost Management and Deployment Tools

Azure Pricing Factors, Pricing Calculator, Cost Management Tools, Budgets and Alerts, Service Level Agreements, Azure Portal, Azure CLI, Azure PowerShell, Cloud Shell, Azure Resource Manager, Infrastructure as Code Principles

TEXTBOOKS/REFERENCE BOOKS

TEXTBOOKS

- **T1:** Exam Ref AZ-900 Microsoft Azure Fundamentals – Jim Cheshire
- **T2:** Microsoft Learn: AZ-900 Study Guide (Official Microsoft Documentation)

REFERENCE BOOKS

- **R1:** Microsoft Azure Fundamentals – Marczak & Nocentino
- **R2:** Azure for Architects – Ritesh Modi

PRACTICALS

Experiment 1: Create a free Azure account and explore Azure Portal, resource groups and subscriptions.

Experiment 2: Compare IaaS, PaaS and SaaS by identifying real Azure services and documenting use cases.

Experiment 3: Create and configure a Virtual Machine and connect using RDP/SSH.

Experiment 4: Create a Virtual Network with subnets and configure Network Security Group rules.

Experiment 5: Create a Storage Account and upload data to Blob Storage.

Experiment 6: Deploy an Azure App Service and host a simple web application.

Experiment 7: Configure Microsoft Entra ID users and assign role-based access control permissions.

Experiment 8: Use Azure Pricing Calculator to estimate deployment cost for a web application.

Experiment 9: Create and apply Azure Policy to restrict resource deployment.

Experiment 10: Deploy a resource using Azure CLI or Cloud Shell and verify deployment.